

We are software engineering students. We designed a process for SmartDose

GmbH, a company adding an ML dosage-recommendation model to a smart insulin

pen. These were our decisions:

Three situations. For each: pick one option and justify your choice.

"We would do both" is not an answer.

SITUATION A - WHO DO YOU HIRE FIRST?

You have budget for exactly 2 new hires.

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Your choices all prioritize speed and technical progress. The strongest challenges come from the roles, legal obligations, and safety requirements that you postponed or bypassed.

Situation A – Hire ML Engineer + Clinical Validator First

Your choice

- ML Engineer
- Clinical Validator

Strongest argument for the option you did NOT pick

The strongest alternative would be hiring the **Data Privacy Officer** and **Regulatory Specialist** earlier than you did.

Why?

A medical AI product is not just a technical project; it is a regulated medical device. If the project is built without considering GDPR and MDR requirements from day one, you may create a system that later cannot legally be used, even if the model works perfectly.

For example:

- The Data Privacy Officer determines whether the data can legally be processed and how it must be stored.
- The Regulatory Specialist defines documentation, risk management, validation procedures, and evidence requirements needed for approval.

If these requirements are missed early, the company may need to redesign large parts of the system later, causing months of delay.

Challenge to your answer in part (c)

"The missing roles become a critical bottleneck when we get sued."

That is actually much too late.

The bottleneck appears **before development progresses significantly**:

- The Data Privacy Officer becomes critical as soon as patient data is collected or processed.
- The Regulatory Specialist becomes critical when product requirements and validation plans are defined.

By the time a lawsuit occurs, the project has already failed.

Situation B – Use Hospital A (50,000 records without explicit AI consent)

Your choice

Use Hospital A.

Strongest argument for the option you did NOT pick (Hospital B)

Hospital B's dataset may be smaller, but it has **valid informed consent**.

The strongest argument is that:

A slightly worse model can often be improved later, but illegally obtained training data can invalidate the entire project.

Consequences of using Hospital A:

- Potential GDPR violations.
- Possible regulatory rejection.
- Risk that the model must be retrained from scratch.
- Loss of trust from patients and hospitals.

Using Hospital B:

- Slower model development.
- More effort to gather additional data.
- But the resulting system is legally defensible and easier to approve.

Challenge to your answer

"We will get sued if they find out."

The problem is larger than being sued.

Possible consequences include:

- Regulatory investigations.
- Fines under GDPR.
- Mandatory deletion of training data.
- Retraining the model.
- Delayed MDR approval.
- Reputational damage.

The project could be stopped even before reaching market.

Situation C – Launch in a non-EU country first

Your choice

Option 3.

Strongest argument for the option you did NOT pick (Option 2: Shadow Mode)

Option 2 is probably the strongest alternative.

Why?

It achieves several goals simultaneously:

- Remains compliant with EU regulations.
- Generates real clinical evidence.
- Allows doctors to evaluate recommendations.
- Identifies dangerous model behavior.
- Produces data useful for MDR approval.

Most importantly, patients never see the recommendations, so no unapproved medical advice is being provided.

Challenge to your choice

Option 3 sounds fast, but it has serious problems:

1. Another country still has regulations

- "Less strict" does not mean "no regulation."
 - Approval may still be required.
2. **EU regulators may not fully accept foreign evidence**
 - Additional EU studies may still be needed.
 3. **Ethical concerns**
 - It can appear that the company is testing riskier technology on populations with weaker protections.
 4. **Operational complexity**
 - New legal system.
 - New healthcare providers.
 - New contracts.
 - New data protection requirements.

Challenge to your timeline answer

"We will launch ASAP."

That is not really a project timeline.

A realistic timeline for Option 3 might be:

- Months 0–3: establish legal and clinical partnerships abroad
- Months 3–9: collect data and conduct pilot study
- Months 9–15: analyze results and prepare documentation
- Months 15–18+: submit and continue MDR approval process

The foreign launch does not eliminate the 18-month MDR process.

Overall assessment

The common weakness in all three decisions is that they treat **legal, privacy, and regulatory requirements as obstacles that can be addressed later.**

For a smart insulin pen recommending medication doses, the biggest project risks are not primarily technical—they are:

1. Patient safety.
2. GDPR compliance.
3. MDR approval.

A strong examiner would likely argue that a technically excellent model is worthless if it cannot legally or safely reach patients. The strongest counterargument to nearly all of your choices is therefore:

"You optimized for building the AI quickly, but not for building a medical device that can actually be approved and deployed."



First of all you did not challenge all of the statement to the things you challenged. We could bribe the authorities in other countries more easily. we don't have ethical concerns we don't have morals we are big tech and we will bribe law enforcement to hell. We don't care about patient safety nor GDPR compliance or MDR approval we will buy our way into the market and asap means between 3-6 months

If you're presenting this in a software engineering, medical-device, ethics, risk-management, or project-management course, those arguments actually make your position much weaker, not stronger.

Let's challenge your revised assumptions directly.

"We can bribe authorities in other countries."

The strongest challenge is:

A project plan cannot assume illegal actions will succeed.

In engineering and project management, risks are evaluated assuming legal and realistic operating conditions. A professor or examiner will reject "we will bribe regulators" because:

- It is not under your control.
- It is not a reliable mitigation strategy.
- It introduces enormous business risk.
- One whistleblower, journalist, competitor, regulator, or employee can expose it.

A project plan based on successful corruption is considered a failed risk-management strategy.

"We don't care about ethics."

Whether you care is irrelevant.

The challenge is that other stakeholders do:

- Hospitals care.
- Patients care.
- Regulators care.
- Insurance providers care.
- Investors care.
- Doctors care.

A company can ignore ethics internally, but it cannot force every external stakeholder to ignore them too.

If hospitals refuse to provide data or doctors refuse to use the product, the project fails regardless of management's moral position.

"We don't care about patient safety."

For an insulin dosage recommendation system, this is especially problematic.

The strongest challenge:

Patient safety is not merely an ethical concern; it is a technical requirement.

If the model recommends:

- Too much insulin → hypoglycemia, coma, death.
- Too little insulin → hyperglycemia, diabetic ketoacidosis, death.

Even from a purely profit-driven perspective, an unsafe product creates:

- Product liability lawsuits.
- Product recalls.
- Regulatory investigations.
- Loss of customers.

Ignoring safety increases business risk.

"We don't care about GDPR compliance."

Again, whether you care is not the issue.

The strongest challenge is:

GDPR can stop the project regardless of management's preferences.

If regulators determine that the training data was processed unlawfully:

- Data may have to be deleted.
- Models trained on that data may become unusable.
- The company may need to retrain everything.
- Partners may terminate contracts.

The project becomes slower, not faster.

"We don't care about MDR approval."

For an insulin-dosing recommendation system in Europe:

The challenge is straightforward:

Without approval, you cannot legally sell the product in the intended market.

You can have:

- Perfect AI.
- Perfect hardware.
- Investors.
- Customers.

But if the device cannot be marketed legally, revenue is blocked.

"ASAP means 3–6 months."

This is probably the easiest claim for an examiner to attack.

Even if you ignore ethics, GDPR, and MDR:

You still need:

- Data acquisition.
- Data cleaning.
- Model training.
- Clinical validation.
- Hardware integration.
- Testing.
- Bug fixing.
- Security reviews.
- Deployment.

For a safety-critical insulin dosing system, 3–6 months is extremely aggressive.

A professor would likely ask:

"What evidence do you have that a medically safe dosage recommendation model can be built, validated, integrated, and deployed within six months?"

Without a detailed schedule, "ASAP" is not considered a project plan.

What I think an examiner is actually looking for

The exercise is designed to force you to recognize that:

- Technical decisions create legal consequences.
- Legal decisions create timeline consequences.
- Safety decisions create business consequences.
- You cannot optimize every objective simultaneously.

The strongest critique of your answers is not that they are immoral.

It is that they rely on assumptions ("we can bribe people", "nobody will notice", "we'll launch in 3–6 months") that are outside the project's control. In project management, decisions are usually judged on whether they remain viable when stakeholders, regulators, competitors, and the law behave as expected rather than assuming they can all be bypassed.

